

real root, and there are only finitely many isomorphism classes of such V . This proves finiteness of the number of $G_{\hat{\mathbf{d}}}$ -orbits in $R_{\hat{\mathbf{d}}}^{\text{flag}}(Q)$. $\square$

Corollary 3.4. *For $\mathbf{d} \in \mathbb{N}Q_0$, the representation space $R_{\mathbf{d}}(Q)$ is spherical if and only if there is no gentle imaginary root $\mathbf{e} \leq \hat{\mathbf{d}}$ for $Q_{\mathbf{d}}$.*

Proof. Sphericity of $R_{\mathbf{d}}(Q)$ is equivalent to $B_{\mathbf{d}}$ acting with finitely many orbits on $R_{\mathbf{d}}$ by Theorem 2.3. This is equivalent to $G_{\hat{\mathbf{d}}}$ acting with finitely many orbits on $R_{\hat{\mathbf{d}}}^{\text{flag}}(Q_{\mathbf{d}})$, and the previous proposition shows the claim. $\square$

4. CLASSIFICATION

To approach the classification of spherical representation varieties, we first show that we can restrict to excluding minimal non-spherical situations.

Lemma 4.1. *If $\mathbf{d} \leq \mathbf{e}$ are dimension vectors for Q and $R_{\mathbf{d}}(Q)$ is not spherical, then $R_{\mathbf{e}}(Q)$ is not spherical.*

Proof. We have an embedding of quivers

$$\iota : Q_{\mathbf{d}} \rightarrow Q_{\mathbf{e}}, \quad (i, k) \mapsto (i, k + e_i - d_i)$$

for $i \in Q_0$ and $k \leq d_i$. It induces an embedding of Weyl groups

$$\iota : W_{Q_{\mathbf{d}}} \rightarrow W_{Q_{\mathbf{e}}}$$

and a W_Q -equivariant embedding

$$\iota : \mathbb{Z}(Q_{\mathbf{d}})_0 \rightarrow \mathbb{Z}(Q_{\mathbf{e}})_0.$$

It is easily verified that ι maps $F_{Q_{\mathbf{d}}}$ to $F_{Q_{\mathbf{e}}}$, and thus imaginary roots for $Q_{\mathbf{d}}$ to imaginary roots for $Q_{\mathbf{e}}$. Obviously, ι also preserves gentleness. The claim then follows from the previous corollary. $\square$

We say that a quiver Q is a *connected sum* of two full subquivers Q' and Q'' along a vertex $i \in Q_0$ if

$$(Q')_0 \cup (Q'')_0 = Q_0 \text{ and } (Q')_0 \cap (Q'')_0 = \{i\}.$$

In this case, we write

$$Q = Q' \#_i Q''.$$

Given a dimension vector $\mathbf{d}$ for Q , we write

$$(Q, \mathbf{d}) = (Q', \mathbf{d}|_{Q'}) \#_i (Q'', \mathbf{d}|_{Q''}).$$

We call such a connected sum of quiver settings *thin* if $d_i = 1$. In the following, the symbol $\leftrightarrow$ denotes an arrow of either orientation.

Theorem 4.2. *For a connected quiver setting $(Q, \mathbf{d})$, the representation space $R_{\mathbf{d}}(Q)$ is a spherical variety under the action of $B_{\mathbf{d}} \subset G_{\mathbf{d}}$ if and only if $(Q, \mathbf{d})$ is a thin connect sum of quiver settings with Q of the form $1 \leftrightarrow 2 \leftrightarrow 3$, and $\mathbf{d} = (m, n, 1)$ or $\mathbf{d} = (m, 2, n)$.*

Proof. First assume $R_{\mathbf{d}}(Q)$ is spherical. We reduce to the claimed situation in several steps, always using the previous lemma.

- (1) If Q contains a cycle, then $R_{\mathbf{d}}(Q)$ even admits infinitely many $G_{\mathbf{d}}$ -orbits, since there are already infinitely many isomorphism classes of representations of dimension vector $(1, \dots, 1)$ for any cycle quiver. Thus we can assume Q to be a tree.